Above operations are necessary to remove oil cooler element; if combination adapter, strainer, and filter are to be removed, then continue as follows:

6. Remove filter and strainer. Remove cap screws that hold adapter to cylinder block then remove adapter.

TRUCK ENGINE

1. Drain cooling system.
2. Disconnect water connection from radiator at lower end of oil cooler.
3. Break hose clamp connection between water pump and oil cooler.
4. Remove cap screws retaining oil cooler cover casting to adapter and remove cover. Be careful when withdrawing assembly not to drop and damage cooling element.

The above operations are necessary to remove oil cooler element; if adapter is to be removed continue as follows:

5. Remove cap screws retaining adapter to cylinder block and remove adapter.

CLEANING

Since the function of the oil cooler is to reduce the engine oil temperature, the oil passages and the water side of the cooler must be kept clean for efficient cooling. The frequency of cleaning will vary depending on the type of lubricating oil used, operating conditions, and the filter medium employed.

When the cooler becomes clogged, the oil pressure on the inlet side of the cooler increases and the outlet pressure decreases, thus causing an increase in the pressure drop across the cooler. When this pressure drop reaches a predetermined value, the oil cooler by-pass valve opens and the oil circulates through the oil cooler by-pass and consequently is not cooled. The gallery oil pressure then becomes less than normal because of the lowered viscosity of the oil due to lack of cooling and the resulting increase in oil temperature.

CLEANING EQUIPMENT

The recommended oil cooler cleaning equipment consists of a gear or centrifugal pump of 25 to 30 gallons per minute capacity and two metal tanks which must be of sufficient capacity to clean the unit or units involved. One tank should contain kerosene, or an equivalent solvent, and the second tank should contain an alkaline base solution, made up in accordance with supplier's recommendations.

CLEANING OPERATIONS

The kerosene, or equivalent solvent, should be circulated through the cooler. If a circulating pump is not available the cooler can be immersed in the solvent for an indefinite period. This will remove some of the dirty oil and sludge and will avoid contamination of the alkaline base solution which is used later to further clean the oil passages. If the cooler is not cleaned immediately after removal from the engine, it should be kept in the solvent solution in order to avoid drying out and hardening of the oil sludge.

The alkaline base solution should be circulated through the oil cooler passages in reverse direction to the normal oil flow. The cooler should be connected with the cleaning solution inlet at the bottom in order to assure that the cooler is filled with the cleaning solution at all times. The cleaning operation usually requires from 20 to 30 minutes, depending on the condition of the unit involved.

Following this operation, the cooler must be thoroughly flushed with clean water and blown dry. If the cooler is not to be reinstalled immediately, some protective measure against rust must be taken.

If the water side of the cooler is coated with mineral deposits, it can be cleaned in a solution of two or three parts muriatic acid to seven or eight parts water, taking care to plug the oil inlet and outlet during this operation in order to avoid the introduction of acid into the oil passages. After the action of the acid on the scale has ceased, the cooler should be thoroughly washed in a neutralizing solution such as ordinary baking soda or soda ash.

REASSEMBLY AND MOUNTING

The lubricating oil cooler is assembled and mounted by reversing the sequence of operations for removal.

1. Clean the old gaskets from the bosses where the oil cooler adapter sets against the crankcase, and affix new gaskets to these bosses.
2. Locate the adapter into place and attach to cylinder block with the cap screws.
3. Clean the old gasket from both sides of the flange on the cooler element, also from the finished face of the adapter and the finished face of the oil cooler cover; then install a new gasket to each flange of the cooler element.
4. Position the cooler element into inside the